

1. How many possibilities are there when a 5-card hand is dealt from a pack of 52.
2. How many possibilities are there when 4 cards are dealt from a pack of 52, given that one of them is the ace of hearts.
3. A theatre company consisting of 6 players is to be chosen from 15 actors. How many possibilities are there if the company must include the most pushy actor?
4. An England cricket touring party consists of 17 members. In how many ways can 11 players be selected for a match:
 - (a) if Joffra Archer must be selected?
 - (b) if Joffra Archer or Ben Stokes must be selected, but not both?
5. Simplify/factorise:
 - (i) $\frac{(n-1)!}{n!}$
 - (ii) $\frac{(n+3)!}{(n+1)!}$
 - (iii) $n! + (n+1)!$
 - (iv) $\frac{(n-1)!}{3!(n-4)!} + \frac{(n-1)!}{4!(n-5)!}$
6. How many possible results are there when 16 people are divided into two groups:
 - (a) one of ten and one of six?
 - (b) each of eight?
7. Find the number of possible results when 10 children are divided into two groups of 5, if the two youngest children:
 - (a) must be in the same group;
 - (b) must not be in the same group.
8. How many possible results are there when 12 people are divided into three groups:
 - (a) of 3, 4, and 5?
 - (b) each of 4?
9. How many ways are there to split 12 people into two groups? Note: the groups do not have to be the same size, and you can have a group of size one.
10. A committee of three is to be chosen from 5 Sixth Formers and 4 Removes.
 - (a) How many possible committees are there?
 - (b) How many of these committees will consist of:
 - (i) one Sixth Formers and two Removes?
 - (ii) two Sixth Formers and one Remove?
11. Four fish are to be chosen to fill an aquarium. The only fish available are 7 piranhas and 9 sharks.
 - (a) How many possibilities are there?
 - (b) For how many of these possibilities will the aquarium contain
 - (i) no sharks?
 - (ii) exactly one shark?
 - (iii) exactly two sharks?
 - (iv) exactly three sharks?
 - (v) four sharks?
 - (c) What is the sum of your answers in Part (b).

- 12.** A committee of five is to be chosen from 6 Sixth Formers and 6 Removes. How many possibilities are there if the committee must contain:
- (a) at least one Remove?
 - (b) at least one Sixth Formers?
 - (c) at least one Remove and at least one Sixth Former?
- 13.** Five different letters are chosen from the letters A, B, C, D, E, F, G, H, and I. Find the number of choices containing:
- (a) no vowels?
 - (b) exactly one vowel?
 - (c) exactly two vowels?
 - (d) all three vowels?
- 14.** How many different integers can be made by multiplying together two or more different digits from 2, 3, 5, and 7?
- 15.** in a five card hand of poker, how many ways are there are being dealt:
- (a) a flush (every card has the same suit)?
 - (b) a full house (three of one number, two of another)?
 - (c) two-pair (two of one number, two of a different number, and any different third number)?
 - (d) three of a kind (three of one number, two additional different numbers)?
 - (e) a straight (all the numbers in sequence, e.g. ace-two-three-four-five, or ten-Jack-Queen-King-Ace)?
- 16.** An ice cream shop offers four scoops for five pounds. You can choose from any eight flavours and do not have to make each choice distinct. How many different choices could you make?
- 17.** A photographic shop has in stock 8 types of colour films and 7 types of black and white. In how many ways could an enthusiast buy 5 types of film, including at least 2 colour and 1 black and white?